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Magnetostratigraphy across the Berriasian–Valanginian stage boundary (Early Cretaceous), at Cehegin (Murcia Province, southern Spain)

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The Berriasian–Valanginian stage boundary near the town of Cehegin in the eastern Subbetic Cordillera of Spain is documented by a detailed ammonite zonation in pelagic limestones. Two magnetostratigraphic sections spanning the uppermost ammonite subzone of the Berriasian and the lower two zones of the Valanginian yielded identical magnetic polarity patterns. Remanent magnetization is predominantly carried by magnetite, and characteristic directions were obtained by thermal demagnetization. The mean characteristic directions from both sites have an inclination of 48°; however, the site declinations are divergent (030° and 074°) due to the tectonic disturbance of the region. The Cehegin polarity pattern can be correlated by means of ammonite and calpionellid zonation to the magnetostratigraphies of the Berriasian stratotype and several Italian sections, thereby enabling a unique correlation to the M-sequence magnetic polarity time scale. The Berriasian–Valanginian stage boundary is in the middle of normal-polarity chron M15n.

1. Introduction

The boundaries between chronostratigraphic stage units of the geological time scale would be best defined by globally isochronous markers. The majority of geological stages have stratotypes. A stratotype is a designated outcrop of a historically documented formation with a unique fossil content. Stage units may be correlated by their component biozones. With increasing stratigraphic resolution, more emphasis has been placed on accurate recognition of the actual boundaries of zones and stages. As a result, a plea has been made to define geological stages by the base of the earliest component biozone at a boundary type section [1]. The base of the biozone is typically defined by the first appearance of a new species or taxon. However, many of these biostratigraphic "datum" levels are not isochronous between basins [2–4] or the taxa may have a limited regional distribution. Correlation between Tethys and Boreal biogeographic provinces of the Late Jurassic and Early Cretaceous is particularly difficult because neither ammonites nor microfossils appear to be sufficiently cosmopolitan to provide isochronous datum horizons [5].

It is therefore important to tie the formal definition of the stage boundary at the stratotype to non-biological events, such as magnetic reversals or recognized worldwide eustatic sealevel changes, which are globally synchronous and recognizable in different sedimentary environments. For example, the Jurassic–Cretaceous period boundary could be defined as the base of magnetic polarity chron M18r [6]. The need for non-biological correlation of stage boundaries has been recognized by the International Stratigraphic Commission—one of the important criteria for selection of the international boundary stratotype section is the suitability for magnetic stratigraphy [1].

2. Berriasian–Valanginian stage boundary

The currently recommended definition of the boundary between the Berriasian (earliest stage of the Cretaceous) and the succeeding Valanginian stage is the base of the ammonite *Thurmanniceras*

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otopeta Zone of the Valanginian [5]. * The underlying *Berriasella callisto* Subzone of the *Fauriella boissieri* Zone therefore represents the uppermost Berriasian. This biostratigraphic definition for the boundary falls within the uppermost subzone, "D3", of the calpionellid *Calpionellopsis* Zone (="Zone D") [8,9].

However, no formal stratotype for the Berriasian–Valanginian boundary has been selected. The candidate sections in southeastern France have been sampled intensively for magnetostratigraphy and were found to be magnetically unsuitable (personal communications of independent unpublished studies by W. Lowrie, by D. Martin, by B. Galbrun and L. Rasplus, and by J. Ogg).

Suitable candidate sections for the boundary stratotype are also located in the vicinity of Caravaca and Cehegin (Province of Murcia) in the Subbetic Cordillera of southeastern Spain. These pelagic carbonate sections are richly fossiliferous, and the ammonite and calpionellid biostratigraphy of the Berriasian–Valanginian has been studied in detail [10–14].

The magnetostratigraphy of two sections across the Berriasian–Valanginian boundary near the town of Cehegin has yielded a reproducible magnetic polarity sequence, as summarized in this report. The Berriasian–Valanginian stage boundary, as defined by the base of the *Th. otopeta* Zone, is in the middle of magnetic polarity chron M15n.

3. Previous magnetostratigraphic studies of the Berriasian–earliest Valanginian

The magnetic polarity time scale for the Early Cretaceous is generally considered to be similar to the marine magnetic anomaly "M-sequence", which is best based on the polarity block model of the Hawaiian magnetic lineations [15,16]. Nomenclature for polarity chrons of the time scale and for the corresponding polarity zones in stratigraphic sections are from the numbered reversed-polarity magnetic anomalies. Thus, reversed-polarity zone M13r in the stratigraphic record was formed during polarity chron M13r and is also represented by marine magnetic anomaly M13. **

Portions of the M-sequence polarity pattern have been correlated to various biostratigraphic zones and datum horizons in numerous sections; these magnetostratigraphic studies are summarized in several recent reviews [4,6,20–24]. Several of these studies are particularly relevant to the bio-magnetostratigraphic correlation of the Berriasian–Valanginian boundary:

(1) Ammonite zones and subzones of the Berriasian stratotype section (Berrias, France) have been correlated to polarity chrons M17r through M15r; the base of the *B. callisto* Subzone (highest subzone of the Berriasian) occurs within polarity chron M15r [25,26]. The Berriasian–Valanginian boundary is not clearly exposed or continuous in this section. No section of Valanginian strata containing index ammonites has yielded useful magnetostratigraphic results.

(2) Calpionellid zones of the Berriasian and lower Valanginian (Calpionellid Zones B through E) have been correlated to polarity chrons M18 through M12 at several sections in central and northern Italy [24,25,27,28] and at the Berriasian section of Berrias [25]. The base of Calpionellid Subzone D3, defined by the lowest appearance of *Lorenziella hungarica*, may be within polarity chron M15n, but this event is poorly represented in the Italian sections. At the Berrias stratotype, the base of Calpionellid Subzone D3 is within polarity zone M15r.

(3) Dinoflagellate datum levels of the Berriasian and earliest Valanginian have been correlated to polarity chrons M19 through M15 at DSDP sites in the western Central Atlantic [23,29]. The Berriasian–Valanginian boundary, as tentatively marked by the first appearance datum of *Druggidium apicopaucicum* [30], appears to occur within polarity chron M15n.

(4) Nannofossil datums of the Berriasian and Valanginian have been correlated to polarity chrons M18 to M10N in Italian sections [28,31,32] and in the western Central Atlantic [23,29]. The

* Ammonite zones and subzones for the Valanginian have not yet been formally standardized. The zonation of Busnardo and Thieuloy [7] is used in this study.

** Terminology for magnetostratigraphic units follows the International and the North American stratigraphic codes [17,18]. Nomenclature for polarity chrons of the M-sequence is that of Harland et al. [19]. A suffix of "r" or "n" denotes the reversed- or the succeeding normal-polarity interval, respectively, of the polarity chron.

Berriasian–Valanginian boundary is considered to lie above the last appearance datum of *Percivalia fenestrata* and below the last appearance datum of *Calcicalathina oblongata* [33], and appears to occur near the polarity chron boundary M15n to M14r in Italian sections [28].

These correlations of various biostratigraphic markers to polarity chrons constrain the traditional Berriasian–Valanginian boundary to occur within the interval of polarity chron M15r to M14r. However, the Berriasian–Valanginian boundary as *defined* by the primary standard of ammonite zonation standard has not been correlated. Considering that the boundary stratotype has not yet been selected, it is not possible to assign the Berriasian–Valanginian boundary to a specific polarity chron.

4. Berriasian–Valanginian boundary sections near Cehegin, Spain

The magnetostratigraphic sections across the Berriasian–Valanginian boundary are exposed along the dry wash, Barranco de Cañada Lengua, located approximately 3.5 km south of the center of Cehegian in the province of Murcia (38.07° N, 1.81° W). Cehegin #1 (VCY) is approximately 700 m upstream from the mouth of this wash into Rio Quipar and Cehegin #2 (VCZ) is approximately 300 m further upstream (Fig. 1).

The sections are within a pelagic basinal facies of the Loma de Solana tectonic unit of the Subbetic Cordillera [11,34–36]. Pelagic sedimentation in this region persisted from the Bajocian (Middle Jurassic) through the Eocene, and the Berriasian through upper Hauterivian strata have no observed hiatuses in sedimentation. Detailed ammonite, nannofossil and calpionellid successions from these Berriasian–Valanginian outcrops of the Barranco de Cañada Lengua have contributed to the development of the standard biostratigraphic zones and subzones for these stages [10–14] and to the current definition of the stage boundary [5].

Magnetostratigraphy sampling covered: (1) the uppermost portion of a red to tan nodular limestone member at the top of the Tollo Formation, (2) a transitional interval of gray to tannish gray marl with thin interbedded marly limestone, and (3) medium-bedded, light gray micritic limestones of the basal Miravetes Formation. The two sections, VCY and VCZ, dip to the southeast by 11 to 18° and 11°, respectively, and have no internal structural complications; however, the sections are separated by a complex shear and contorted tectonic zone. The lower boundaries of the magnetostratigraphic sections are fault truncations.

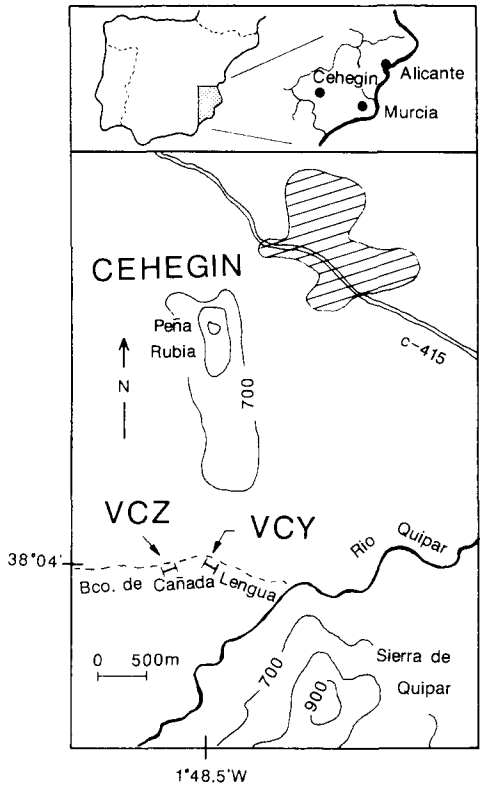


Fig. 1. Location of the magnetostratigraphic sections (VCY and VCZ) near Cehegian in the Province of Murcia, southern Spain.

Ammonites of these two sections have been sampled on a bed-by-bed basis adding details to previous paleontological investigations [14]. The base of the *Thurmanniceras otopeta* Zone, the recommended Berriasian–Valanginian boundary, is defined by the lowest occurrence of the index ammonite *Th. otopeta* in a set of marly limestone beds; the precision in the placement of this datum is the thickness of the sampled bed. The overlying transition into the *Th. pertransiens* Zone is defined by the lowest occurrence of the index ammonite *Th. pertransiens*; in section VCZ this datum also appears to be coincident with the boundary between Calpionellid Zones D and E [14] as has

been observed in southeastern France [37]. No calpionellid zonation has been reported from section VCY, and the calpionellid identifications in section VCZ do not allow subdivision of Zone D into subzones. The uppermost portions of these particular sections did not yield ammonites diagnostic of the succeeding *Busnardoites campylotoxus* Zone, but no detailed collecting has been made.

Local folding and numerous faults have distorted the stratigraphy between and surrounding these sections. Contributing to this local disruption is extensive internal compressional and transverse tectonic deformation of the Subbetic Cordillera. Therefore, the rotations indicated by the paleomagnetic data reflect only local tectonic block movements and cannot be applied to the Subbetic microplate or to the Iberian plate [38,39].

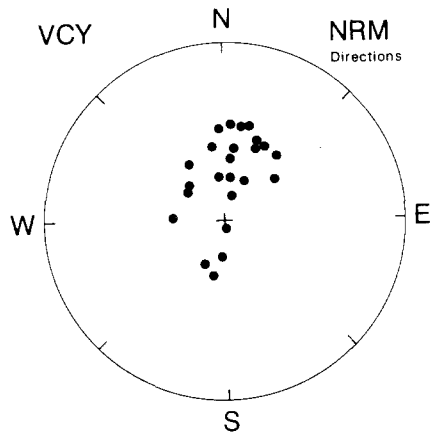


Fig. 2. NRM directions from section VCY; equal area plot, solid symbols are lower hemisphere, open symbols are upper hemisphere. Section VCZ has a similar distribution of NRM directions.

5. Analytical procedure and magnetic properties

Paleomagnetic analyses and demagnetizations of the sample suites (25 minicores from VCY, 48 from VCZ) were performed in a steel-shielded room at the University of Wyoming using a computer-linked ScT cryogenic magnetometer. All samples were treated by progressive thermal demagnetization: natural remanent magnetization (NRM) plus 10 heating steps to 550 °C for the

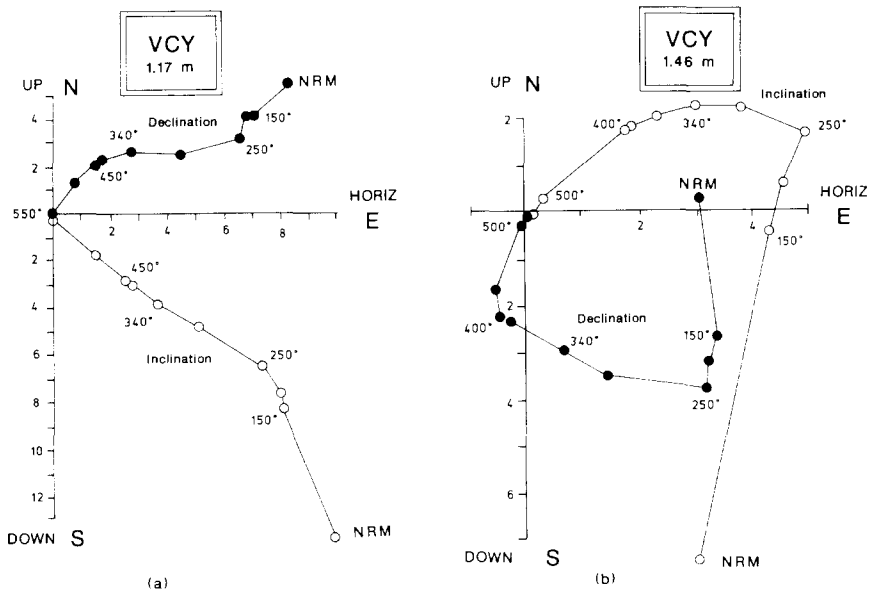


Fig. 3. Vector plots of thermal demagnetization behavior of two stratigraphically-adjacent samples from section VCY (1.17 and 1.46 meter levels for plots to right and left respectively). Declination (solid dots) is the horizontal component of the magnetization vector; inclination (open dots) is the angle from horizontal and the total intensity of the magnetization vector; vectors corrected for tilt of bedding; one scale division is equal to 10^{-4} A/m. Some irregularities in spacing of demagnetization steps were caused by oven problems. For each sample, the characteristic magnetization is the least-squares fit to the four vectors from 370 °C to 500 °C. Polarity assignments are normal (N) polarity and reversed (R) for right and left samples respectively.

VCY suite; NRM plus 8 steps to 500 °C for the VCZ suite. Vector plots of the directions of magnetization for each sample were examined to assign polarity and to identify linear segments indicating removal of various magnetic components of the NRM.

NRM directions, before tilt correction, were clustered near the present axial field (Fig. 2). Intensities of NRM decreased from an average of 3×10^{-3} A/m for the pinkish nodular limestone of the lower *B. callisto* Subzone to an average of about 1×10^{-4} A/m for the light gray micritic limestone of the upper *Th. pertransiens* Zone; these relative differences in intensity with lithology were preserved through demagnetization.

Directions of normal- and reversed-polarity samples diverged upon heating to 150 °C, and characteristic magnetizations having a linear or quasi-linear decay to the origin were generally maintained from 370 ° to 500 °C (Fig. 3a and 3b). The intensities of magnetization tended to decrease by an order of magnitude during heating from the 500 °C step to 550 ° with the directions commonly becoming spurious. This magnetic behavior suggests that the characteristic magnetization is carried by magnetite. Other than the decreasing intensity upsection, there was no significant difference in general magnetic behavior between the lowermost pinkish marly limestone and the uppermost light gray micritic limestone samples.

The direction of characteristic magnetization and associated variances were computed for each sample by applying a three-dimensional least-squares line fit method [40] to selected regions of univectorial decay, which generally included three to four thermal steps. If the standard deviation of the removed vector was 10 ° or greater, or if the sample displayed irregular behavior during demagnetization, the quality of the characteristic magnetization direction was considered to be "poor", and is denoted by a "p" in the magnetostratigraphic plots. The few samples having indeterminate or questionable polarity interpretations are denoted by "INT".

6. Mean directions

Characteristic directions of magnetization for VCY and VCZ are displayed in Fig. 4a and 4b. Mean directions of magnetization for normal-polarity and for reversed-polarity samples of each section were calculated by using a method modified from Fisher [41]. The computational procedure gave half-weight to characteristic directions of poor quality (as explained above), omitted samples of questionable or indeterminate polarity, and, after the first iteration, omitted samples with directions falling outside two standard deviations from the mean. Mean directions were also computed for all samples combined at each site and

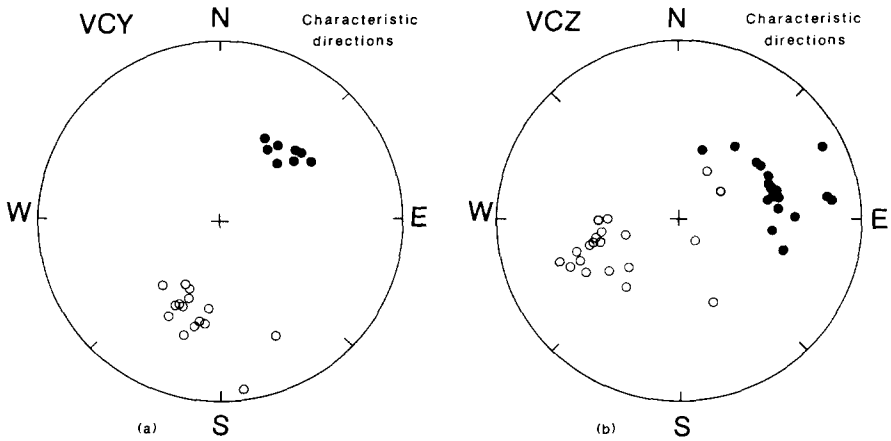


Fig. 4. Characteristic directions of magnetization for sections VCY and VCZ as used in determination of mean directions; equal area plot, solid symbols are lower hemisphere, open symbols are upper hemisphere. Samples having indeterminate polarity are omitted.

TABLE 1

Mean directions of characteristic magnetization (after structural corrections)

Polarity	N	D (°)	I (°)	α_{95} (°)	K	Mean intensity ($\times 10^{-4}$ A/m)	VGP lat. °N	VGP long. °E	dp/dm	Paleolat. (°N)
Section VCY										
N	8	42.8	47.4	(5.6)	100	3.5	53.4	90.3	(4.7/7.2)	28.5
R	13	21.3	46.9	(4.7)	77	0.7	69.7	111.2	(3.9/6.1)	28.1
Comb.	21	29.4	47.6	(4.4)	52	1.3	63.8	100.5	(3.8/5.8)	28.7
N+R	21	32.1	47.2				61.6	98.6		28.3
Section VCZ										
N	22	74.5	45.9	(5.1)	38	2.2	28.0	74.1	(4.2/6.5)	27.3
R	14.5	69.0	50.2	(6.0)	43	0.4	34.0	73.1	(5.4/8.0)	30.9
Comb.	36.5	72.4	47.7	(3.8)	39	1.1	30.3	73.7	(3.2/5.0)	28.8
N+R	36.5	71.8	48.1				31.0	73.4		29.1

N = number of samples (poor quality characteristic directions given half-weight, questionable polarity samples omitted); α_{95} = radius of circle of 95% confidence about mean direction; K = dispersion parameter; mean intensity computed as logarithmic mean; VGP = virtual geomagnetic pole (dp/dm is the oval of confidence)

for the average of the normal- and reversed-polarity directions (Table 1).

The normal- and reversed-polarity directions of VCZ are nearly antipodal. The normal- and reversed-polarity directions of VCY differ in declination by 21°, but have antipodal inclinations; this declination deviation is most noticeable in the lower reddish nodular limestone unit and indicates the persistence of a secondary overprint with an eastward declination.

The mean declinations of the magnetization of the two sites diverge: VCY has a mean declination of 30°, whereas VCZ has 70°. This divergence indicates local block rotation, consistent with observed structural disruption on the local and regional scale.

The mean inclinations of the two sites are an identical 47.5°, implying a paleolatitude of approximately 29°N (±3.5°). For comparison, Tithonian paleolatitudes for the Subbetic region are 20°N (±3°) [38], implying rapid northward drift of the region during the Berriasian. A similar rapid northward drift of about 5° during the Berriasian has been reported from Atlantic DSDP sites [23].

7. Magnetic stratigraphy and assignment of polarity chrons

Magnetic stratigraphies for the two sections, VCY and VCZ, are shown in Figs. 5 and 6, with biostratigraphic (ammonite and calpionellid zones) and lithologic columns. Both sections display a polarity pattern of a basal normal-polarity zone followed by a pair of reversed- and normal-polarity zones of subequal thickness; sampling terminated within an upper, thick, reversed-polarity zone. The lower zone of reversed polarity is documented at VCZ only by pairs of samples at its top and base; the intervening claystone interval was not sampled. The *B. callisto–Th. otopeta* zone boundary, hence the Berriasian–Valanginian stage boundary, lies in the lower half of the middle normal-polarity zone.

Intensity of magnetization decreases upsection with the changes in lithology from red marly limestone to dark gray marl to gray limestone. We interpret this lithologic succession as resulting from a general increase in the rate of sedimentation with an accompanying partial reduction and dilution of iron-rich minerals. The net sedimentation rate of the VCZ (Cehegin #2) section appears to be approximately 50% greater than that of section VCY, especially within the interval of the *Th. otopeta* Zone—that ammonite zone and its associated normal- and reversed-polarity zones are thicker and contain a higher proportion of carbonate. Such variations of thickness and of lithology between the two sections were perhaps caused by irregular bottom topography or by bottom currents.

Correlation of the magnetic polarity columns of

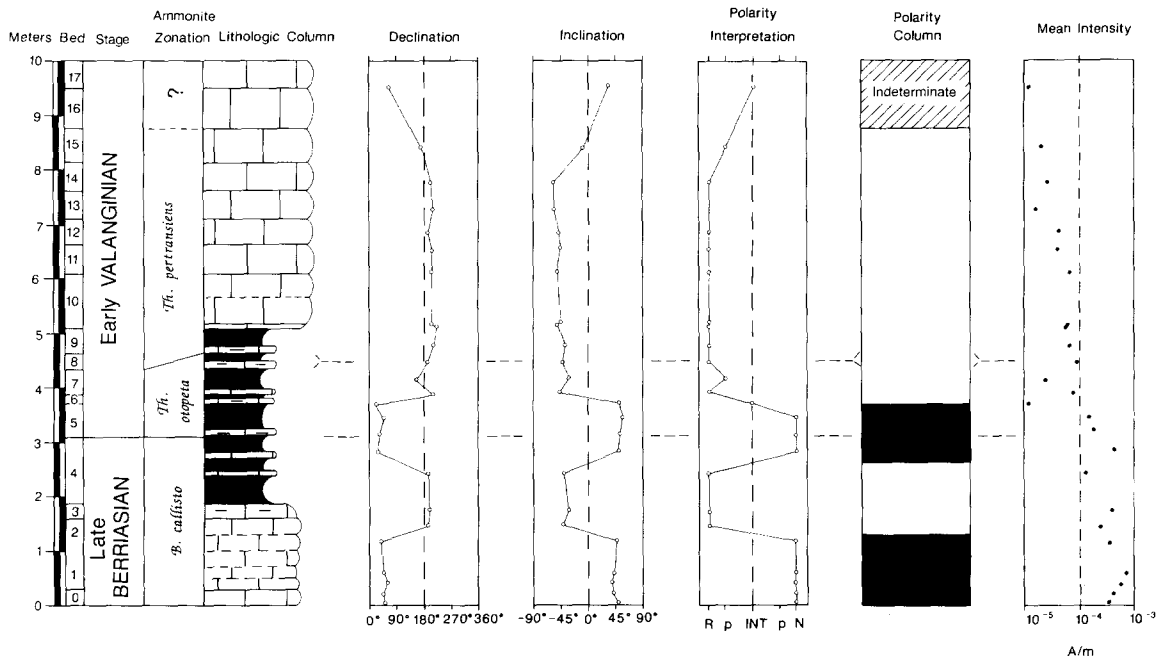


Fig. 5. Stratigraphic plot of characteristic directions and intensities of magnetization for section VCY. In the polarity interpretation column, *N* is normal polarity, *R* is reversed polarity, *p* indicates poor stability, and *INT* indicates indeterminate or questionable polarity interpretation. In the graphical polarity diagram, black is normal polarity, white is reversed polarity and hachured is indeterminate or unsampled.

the two sections is obvious using the biostratigraphic control (Fig. 7). However, this polarity sequence is too short to allow an unambiguous correlation to the M-sequence magnetic polarity time scale. It was necessary to correlate the biostratigraphy and magnetic polarity zones of Cehegin sections to other magnetostratigraphic "reference" sections which have well-defined ammonite or calpionellid zonations, span a longer time interval and have reliable correlations to magnetic polarity chrons of the M-sequence (Fig. 7). In turn, this match allows assignment of polarity chrons to the Cehegian sections.

The primary reference section is the ammonite-zoned Berriasian stratotype [25,26]. In addition, the identification of Calpionellid Zones D and E in section VCZ enable correlation to the detailed calpionellid zones and magnetostratigraphies of three Italian sections: the Bosso section [27] with revised correlation of calpionellid zones to magnetostratigraphy [24] and the Xausa and Capriolo sections [24,28].

The magnetostratigraphic correlation of the Cehegian sections to the reference sections, hence to the M-sequence, is constrained by the ammonite biostratigraphy and the calpionellid zonation (Fig. 7). Each of the Cehegian sections extends from the upper part of polarity zone M16n to the lower part of polarity zone M14r.

The base of the Berriasian *B. callisto* Subzone is not well-documented within either section due to a general lack of ammonites in the lowest beds; however the base must be below the lowest occurrence of *B. callisto* in bed 5 of VCZ, hence within polarity zone M16n. At Berrias, the base of the *B. callisto* Subzone occurs slightly higher—within the lower part of polarity zone M15r.

The base of the *Th. otopeta* Zone, which marks the Berriasian–Valanginian boundary, is in the center of polarity zone M15n.

The base of the *Th. pertransiens* Zone is in middle to lower part of polarity zone M14r, as is the Calpionellid Zone D to E boundary in section VCZ. This latter assignment is similar to the D to

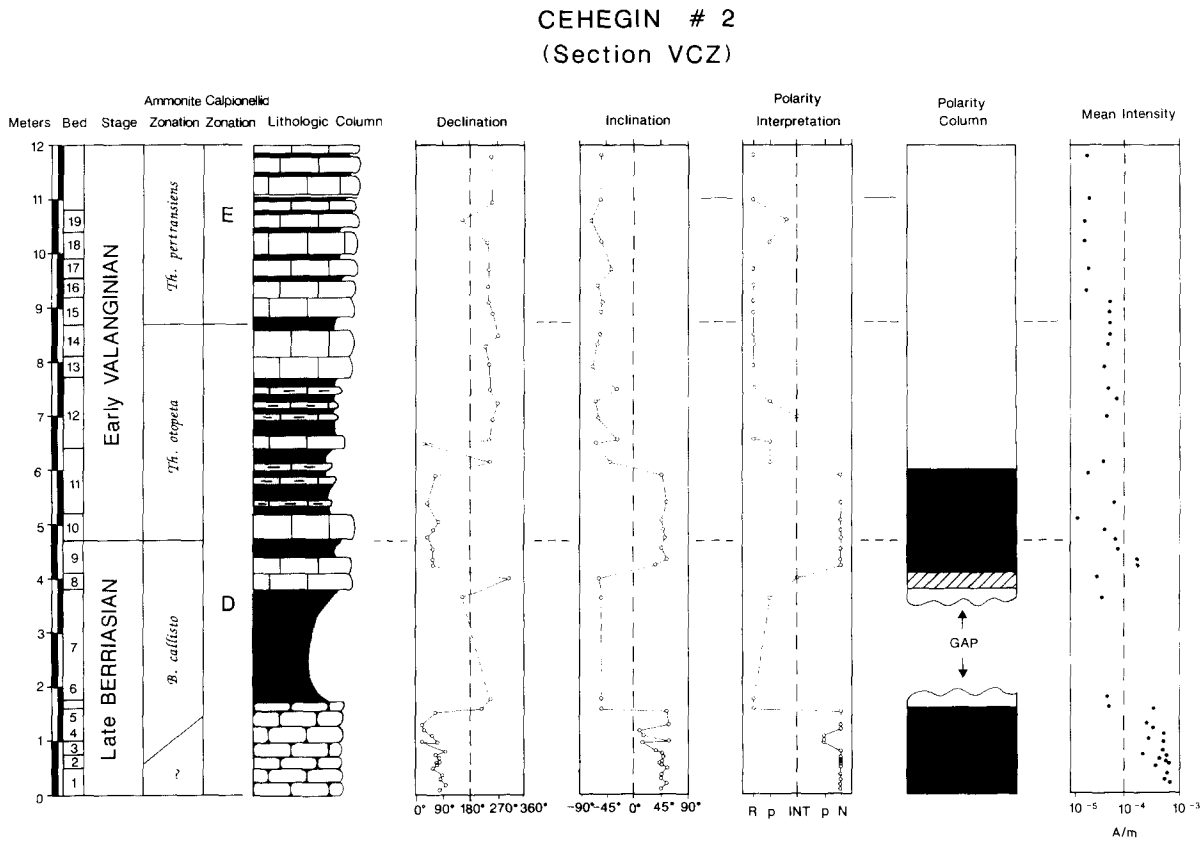


Fig. 6. Stratigraphic plot of characteristic directions and intensities of magnetization for section VCZ. In the polarity interpretation column, *N* is normal polarity, *R* is reversed polarity, *p* indicates poor stability, and *INT* indicates indeterminate or questionable polarity interpretation. In the graphical polarity diagram, black is normal polarity, white is reversed polarity and hachured is indeterminate or unsampled.

E boundary in the Capriolo section; however, this boundary is reported to lie at the top of polarity zone M14r or base of M14n in the other Italian sections [24,28]. This apparent variability in the position of the Zone D to E boundary (and also the C to D boundary) relative to polarity zones in these sections suggests an imprecision in identifying calpionellid zones.

The anomalous thickness of polarity zone M14r in the Cehegian sections is probably the result of the increasing sedimentation rate upsection.

8. Discussion and summary

8.1. Sources of imprecision

The Berriasian–Valanginian boundary in these sections lies in the middle to lower middle of polarity zone M15n. A more precise assignment, in terms of a fixed proportion within the polarity chron, is not possible, because: (1) at each section, the present placement of the *B. callisto–Th. otopeta* ammonite zone boundary has an uncertainty of the thickness of the individual bed or of the marl layer between beds; this limitation is imposed by the method of paleontological collecting; (2) variations in sedimentation rate between and possibly within beds are indicated by fluctuations in the carbonate and clay contents; additionally, there may be diastems or short hiatuses in sedimentation; (3) the characteristic magnetization carried by magnetite is not set at the depositional surface, but a few centimeters below the sediment-water interface [42,43]; therefore, the *absolute* time horizon of the reversal event of polarity chron M15r to M15n would be slightly above the event recorded by magnetic stratigraphy. For these vari-

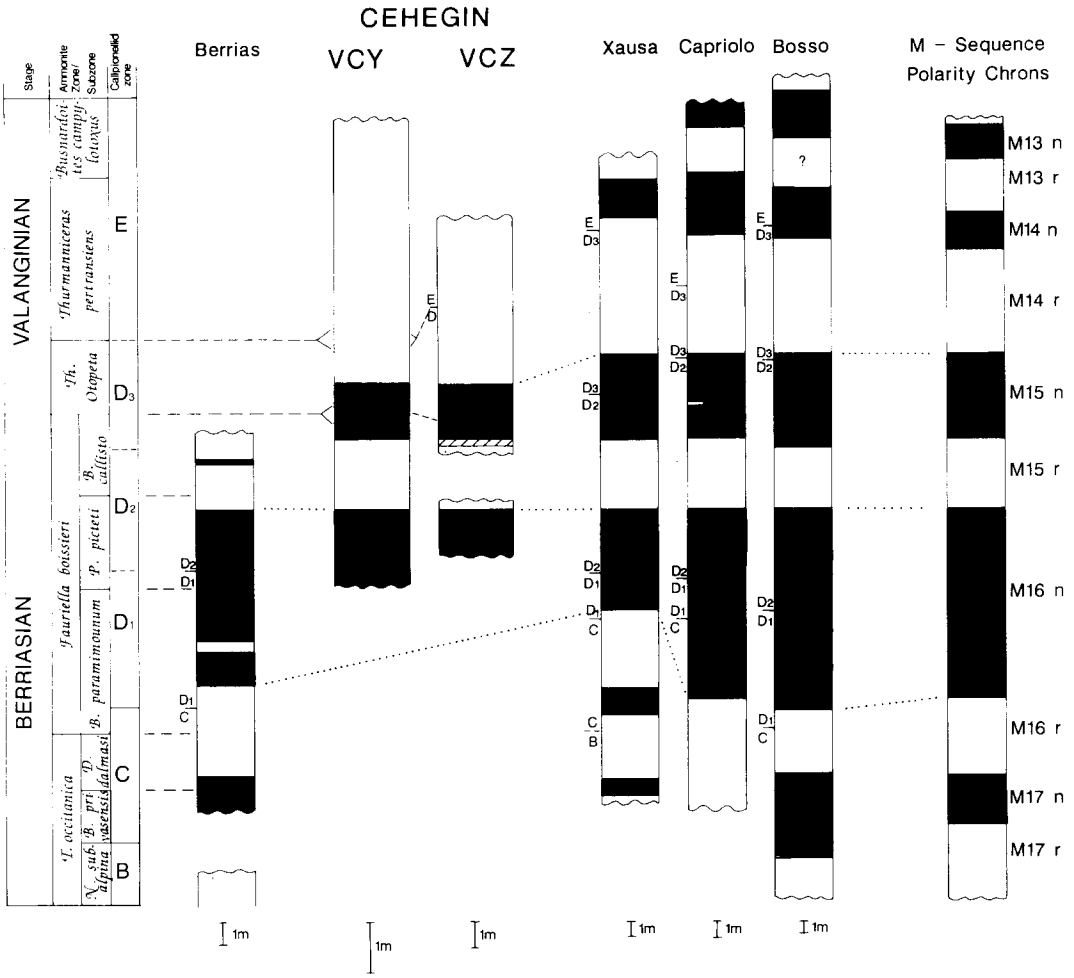


Fig. 7. Correlation of the magnetostratigraphy of sections VCY and VCZ to the magnetostratigraphy from Berrias, France [25,26], from Xausa and Capriolo in northern Italy [24,28], and from Bosso in central Italy [24,27,28] using ammonite and calpionellid biostratigraphy for controls on the correlation. Ammonite zone boundaries are indicated by dashed lines; correlation of polarity zones are indicated by dotted lines. The assignment of polarity chrons from the M-sequence to sections VCY and VCZ is based upon the similarity of the M-sequence pattern to the composite of all sections. (Preprints of the Italian magnetostratigraphies were provided by J.E.T. Channell.)

ous reasons, the precision of the polarity chron assignment is limited to the "middle" of polarity chron M15n, with a possible error of perhaps ± 0.2 m.y. [19]. In the future, it may be possible to narrow this uncertainty, especially if thicker sections are used.

8.2. Global correlation of the boundary

The biostratigraphic marker and definition for the Berriasian–Valanginian boundary in these sections is the lowest occurrence of *Thurmanniceras otopeta*. Because there are no precursors to this taxon or these genera in the sections, this ammonite species possibly arrived by migration. Therefore, the timing of the *Th. otopeta* appearance will probably vary between regions. The French sections also lack an obvious predecessor of *Th. otopeta*, but this may be partially a preservation failure (P.F. Rawson, personal communication, 1987).

A more useful global marker for the Berriasian–Valanginian boundary, one which would not be dependent upon diachronous faunal events, is the magnetic reversal event at the base of polarity

chron M15n (or top of polarity chron M15r). This magnetic reversal precedes the biostratigraphic marker by perhaps 0.3 m.y. The correlation of the Berriasian–Valanginian boundary to the magnetic polarity time scale will enable future correlation to other biogeographic provinces outside of southern Europe in which *Th. otopeta* has not been documented. It is important that a precise magnetostratigraphy be possible in the boundary stratotype section from which it is selected.

The polarity chron placement of the Berriasian–Valanginian boundary, as defined by the ammonite stratigraphy in these sections, is slightly lower (earlier) than previous estimates using correlation of calpionellid or nannofossil markers to magnetostratigraphy. Studies of Italian pelagic carbonates tended to favor placement of this boundary within the upper part of polarity chron M15n or within polarity chron M14r [24,28,31].

8.3. Summary

The Berriasian–Valanginian boundary, as defined by the base of the ammonite *Th. otopeta* Zone in two candidate sections for the international boundary stratotype, is in the center of magnetic normal-polarity zone M15n.

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